3 The Hodgkin-Huxley Equations

There has been nothing of significance in Neuroscience since Hodgkin and Huxley.
—Jerome Y. Lettvin, personal communication, 1976

This book is about model-based tracking and control. All of the neuronal models we will employ are related, directly or indirectly, to the work of Hodgkin and Huxley in the 1940s and 1950s. Although research into *axonology* is no longer viewed as the cutting edge of neuroscience research, the potential pitfalls raised without a deep understanding of these dynamics are very much at the forefront of current work [MSY07]. Later, in chapter 12, we will reexamine the Hodgkin-Huxley equations in a control theory framework. These equations are the foundation of biophysically based models of neuron dynamics.

This will be a long chapter. Few readers will likely ever have much experience in testing in the laboratory the fundamental features of excitable neuronal membranes. With squid axonology largely over, the present-day experience is most often with mammalian neurons of extraordinary spatial complexity. These neurons are typically populated with a *zoo* of ion channels [GK02], whose temporal and spatial dynamics are very complex. At the same time, some of our best textbooks give only brief descriptions of the foundational works that are insufficient to develop intuition. In this chapter, I will take some time to develop a description of the foundational membrane phenomenology enough for the reader to develop a sense for the dynamical intuition behind the Hodgkin-Huxley formalism. All of our biophysically based neuronal models developed in recent decades is commentary and complexity. And understanding these more complex biophysical models is key to our later discussion of Parkinson's disease and epilepsy in chapters 10 and 12.

3.1 Pre-Hodgkin and Huxley

By the turn of the twentieth century, Julius Bernstein had formulated the hypothesis that all living cells were surrounded by a membrane with low permeability to ions [Ber12]. This was a remarkable speculation well before there was direct proof of the existence of such

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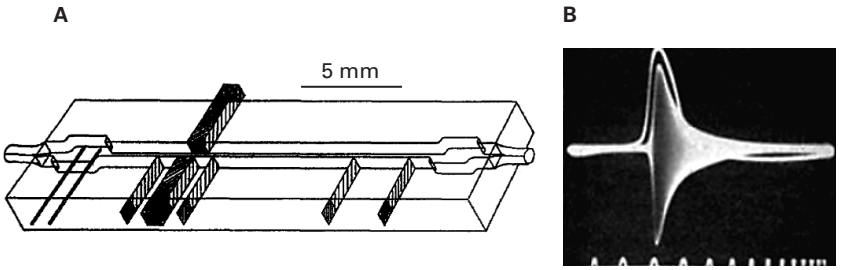


Figure 3.1
Cole and Curtis's apparatus, A, for measuring the impedance change during passage of an action potential, B. In B the action potential and bridge imbalance are shown superimposed. Modified from [CC39] with permission.

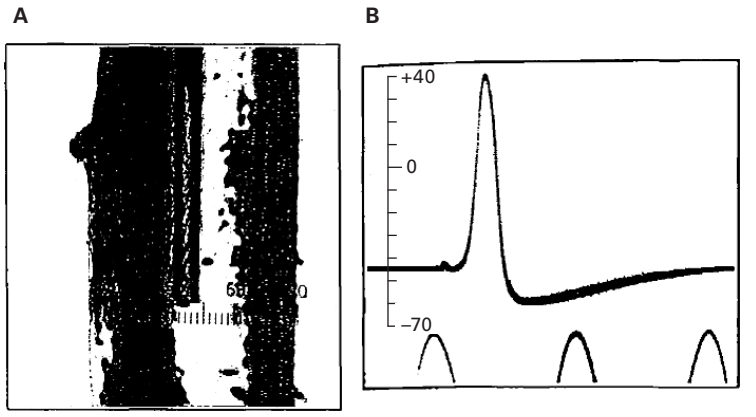


Figure 3.2
The first measurement of overshoot of an action potential. A shows the impalement of the squid giant axon by a glass micropipette. B shows the measured action potential overshoot to near +50 mV with respect to the external bath solution. Reproduced from [HH39] with permission.

structures.[1] Bernstein further hypothesized that there was a resting transmembrane potential gradient, and that action potentials (recorded extracellularly from peripheral nerve) were the result of the breakdown of permeability barriers in the neuronal membrane [Ber12]. This hypothesis predicted that during the action potential, the transmembrane voltage gradient would collapse to near zero.[2]

1. The indirect evidence was strong. Biophysical work with living tissues, especially suspensions of blood cells, demonstrated impedance and reactance characteristics that firmly pointed to electrolyte-containing structures surrounded by a dielectric capacitance [Col68].

2. It would actually collapse to the Donnan equilibrium potential, if impermeant anions remained within the cell [NRB04]

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Cole and Curtis [CC39] performed an ingenious series of transverse measurements of the impedance of a squid giant axon,[3] and demonstrated through imbalance of an alternating-current Wheatstone bridge that during the action potential, a large decrease in impedance occurred consistent with Bernstein's hypothesis (figure 3.1). Unexplained in figure 3.1B was the unusual temporal interplay between the course of the action potential (thin line), and the impedance change (broad envelope). It will take the rest of this chapter to clear this up.

Then came the first direct measurements of transmembrane voltage [HH39]. A significant overshoot of transmembrane potential was observed (figure 3.2). Bernstein's hypothesis was dead.

3.2 Hodgkin and Huxley and Colleagues

As the concept of feedback control electronics matured during the 1940s [Wie48], the ability to devise voltage feedback control systems in biological experiments became feasible. Cole [Col49] and Marmont [Mar49] devised a *voltage clamp*. A key problem was that the voltage of an action potential varied in time as well as space. To eliminate spatial variation, a *space clamp* was employed, whereby the longitudinal (axial) electrode shown in figure 3.3 was used to eliminate potential variations along the length of the axon. To ensure that there were no voltage gradients except across the central region of the axon, they used guard electrodes polarized to the same potential as the central electrode [Mar49] (figure 3.3).

An additional technical need, as Cole[4] introduced [Col68], was to separate the electrode pairs that pass current from the ones that sense the potential changes. Hodgkin, Huxley, and Katz [HHK52] varied Cole's design in order to measure both transmembrane voltage and current, as shown in figure 3.4.

The electronic schematic for this four-electrode technique is illustrated in figure 3.5. We will incorporate this latter principle in the large-scale voltage clamps that we use in neuronal network control experiments described in chapter 11.

The notation used by Hodgkin and Huxley (and Katz) is as follows [HHK52]. V is the displacement of membrane potential, E, from rest, E_r

$$V = E - E_r$$

where E is taken with respect to a distant reference, and sign is by convention outside minus inside. The resting membrane potential (with respect to the fluid outside of a cell) is

3. The description by J. Z. Young of the giant axon of the squid [You36] greatly aided the avalanche of neurophysiological advances over the next sixteen years. The axon was large enough to permit not just impalement with an electrode, but insertion of space clamp electrodes, as well as squeezing out and replacement of the axoplasm.

4. The author is grateful to K. C. Cole for informally reviewing his first manuscript when working with John Moore and Normal Stockbridge, later published as [MSS82]. Such formative encounters inspire preservational chapters.

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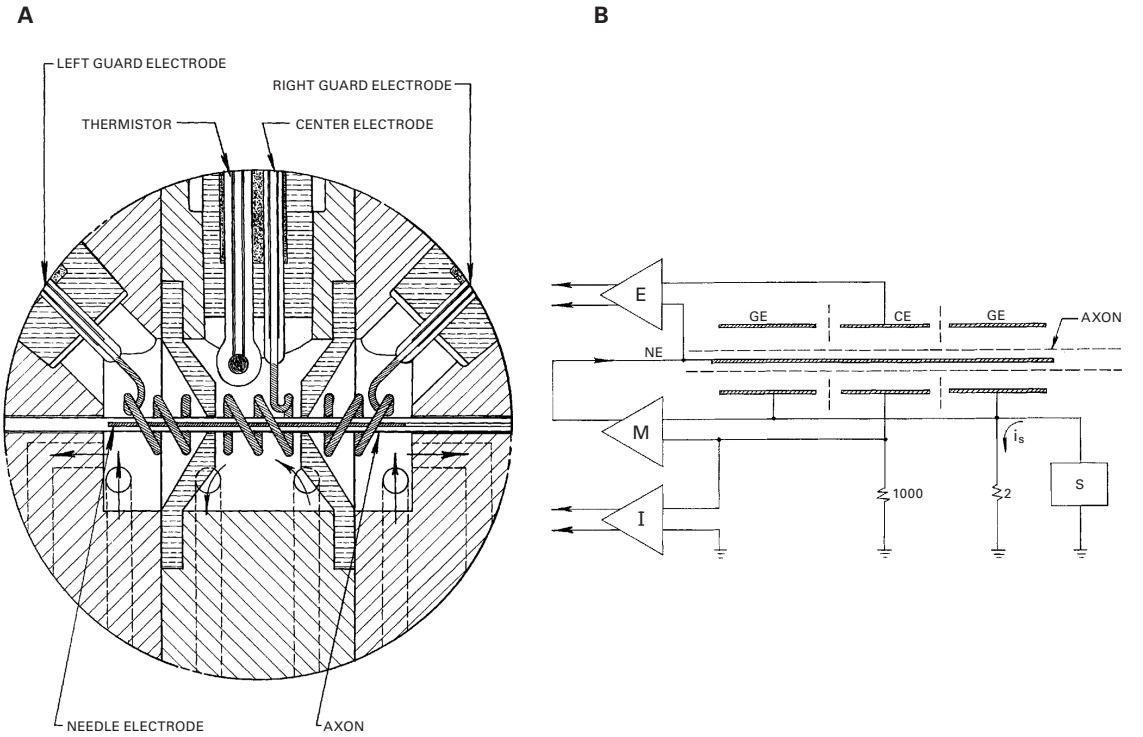


Figure 3.3
Marmont's schematic of his voltage clamp apparatus, A, and electronics, B, that he shared with Cole. Reproduced from [Mar49] with permission.

considered 0 mV. The current sign is positive for positive charge flowing into a cell. They had a strong measurement for the cellular membrane capacitance, C_m, measured from alternating currents passed through suspensions of blood cells measured by Fricke in the 1920s to be 1 μF/cm^2 [Fri24, Fri25]. The *unit membrane* [Rob57], the lipid bilayer of all cells, vertebrate and invertebrate, would turn out to have a universal physical capacitance constant.

We assume that the total membrane current, I_m, is a sum of membrane capacitive current, $C_m \partial V/\partial t$ and membrane ionic current, I_i.

$$I_m = C_m \frac{\partial V}{\partial t} + I_i \tag{3.1}$$

Using voltage clamp feedback to hold the membrane voltage constant, which in equation (3.1) would force the capacitive current to zero, would leave $I_m = I_i$. This was the key to studying the ionic membrane currents, and sorting them out one by one.

The first thing that Hodgkin, Huxley, and Katz [HHK52] observed was the response to membrane potential from brief shocks (figure 3.6). For hyperpolarizing charges, driving

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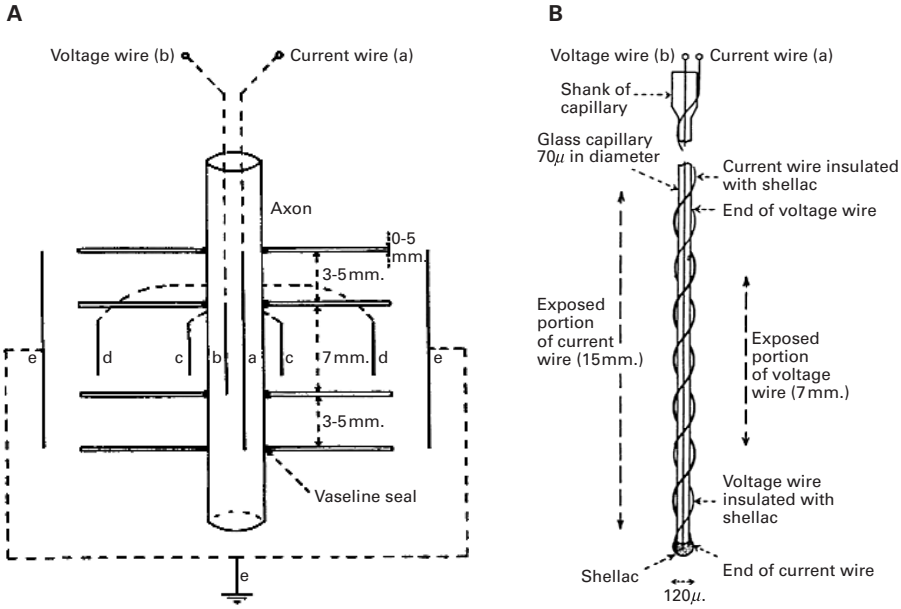


Figure 3.4
Voltage clamp schematic, A, and axial electrode configuration, B, from Hodgkin, Huxley, and Katz [HHK52]. Reproduced from [HHK52] with permission.

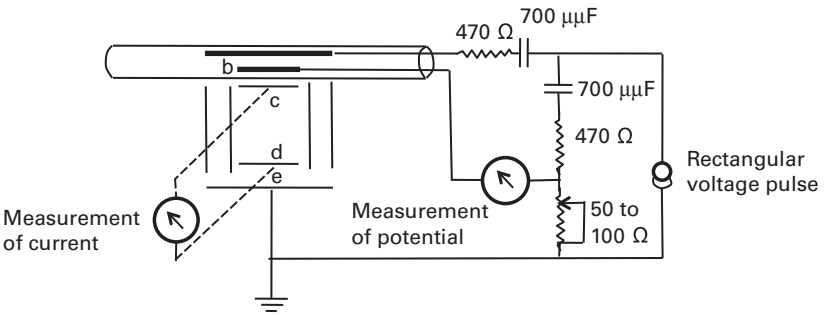


Figure 3.5
Four-electrode technique used to separate current driving and sensing electrode pairs. Reproduced from [HHK52] with permission.

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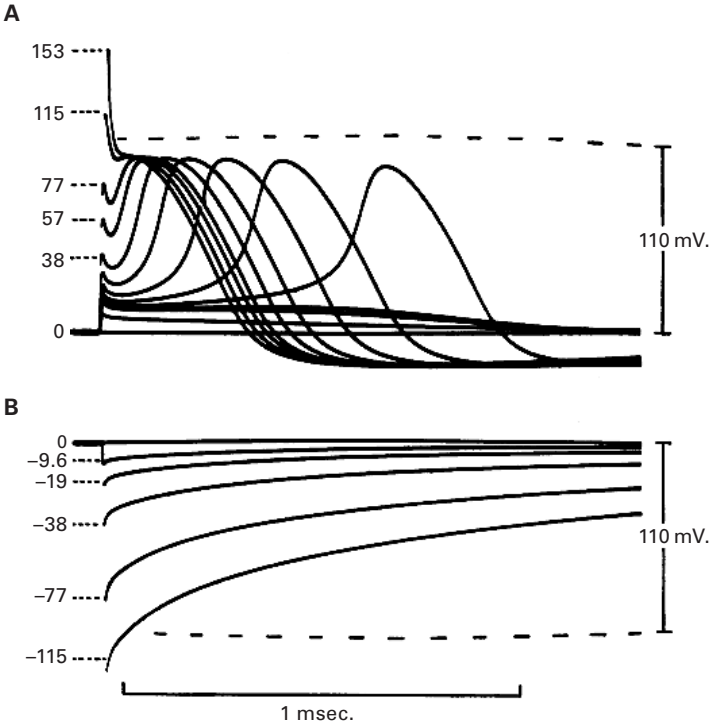


Figure 3.6
Membrane action potentials in response to brief shocks using the apparatus in figure 3.4. Reproduced from [HHK52] with permission.

the inside of the axon more negative than that of rest potential (normally about -70 mV with respect to the outside of the axon), the membrane is charged with the pulse, and then decays like a leaky capacitor (figure 3.6B). For small depolarizing positive shocks, there is similar decay with sign inverted (figure 3.6A). But for slightly larger depolarizing shocks, there is a much larger excursion (figure 3.6A). These *membrane action potentials* were uniform within the central chamber between the guard chambers. They are not propagating, and uniform within the restricted region of the space clamp. Note that, for depolarizing shocks just over the apparent threshold, the delay to the regenerative action potential is longer (figure 3.6A).

Exploring the region of the threshold gave amazing responses (figure 3.7). *If the potential was displaced to the threshold level it might remain in a state of unstable equilibrium for considerable periods of time* [HHK52]. Neural fibers close to threshold are indeterminate in that you can't tell whether they will or won't fire an action potential or when that action potential will occur. They are much more reliable with larger stimuli.

I must say when I first saw these pictures, this indeterminacy appeared to have philosophical implications. It still does. Much has been written about the origins of such concepts of

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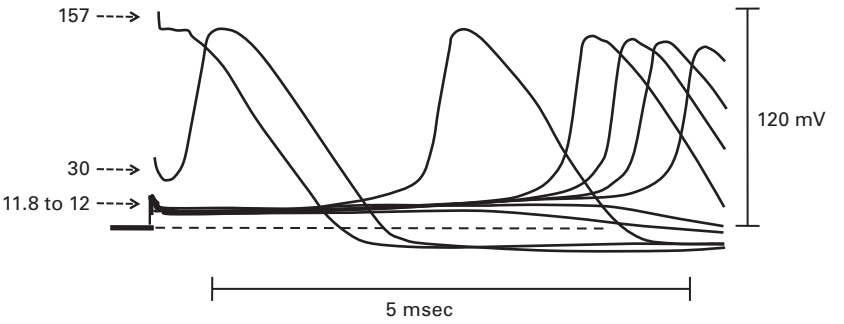


Figure 3.7
Indeterminacy in action potential behavior near threshold. Reproduced from [HHK52] with permission.

free will, which appeared difficult to reconcile with the deterministic formulation of brain dynamics, such as the Hodgkin-Huxley equations we are about to examine. There is a considerable literature, starting with Schrodinger [Sch44b], and promulgated by Eccles [Ecc53] and later Penrose [Pen94], exploring whether quantum indeterminacy needs to be invoked to render a nervous system free from automatism. Missing from the following description of the Hodgkin-Huxley equations is the fundamental stochastic molecular conformational changes that the ion channels exhibit. In later decades, such molecular fluctuations, based on quantum mechanics but only requiring a classical probability description, would be used to explain, first, hemoglobin's binding of oxygen [MWC65] and, later, to account for the indeterminacy as seen here [Fit65] [SFS98].

There was also something deeply wrong with figure 3.6. Look at the plot of voltage delivered and the current that developed 0.29 msec after the shock was delivered (figure 3.8). From rest, a positive (hyperpolarizing), or small negative (depolarizing) deflection yields a linear response. This is Ohm's law. Give a larger negative response, and the response is not just nonlinear. Ohm's law reverses, and, were you to measure between the starting point and the point on the curve for a given depolarizing pulse, the chord conductance [Tho86] would be negative.

Now introduce a voltage clamp. Under voltage clamp, neurons do not fire action potentials (because their voltages are kept from changing). But the electronics in figure 3.5 permit us to change this voltage quickly. In figure 3.9, the upper trace was the current that evolves in response to a +65-mV step hyperpolarization, while the lower trace was in response to a −65-mV depolarization. There is relatively little current that flows for the hyperpolarizing pulse. But for the depolarizing pulse, there is a transient inward current, followed by a prolonged outward current.

Now plot the differences between the current-voltage (I-V) curves for short (0.6 msec) and long (steady state) times as in figure 3.10. The change in current that develops is a function of both voltage and time.

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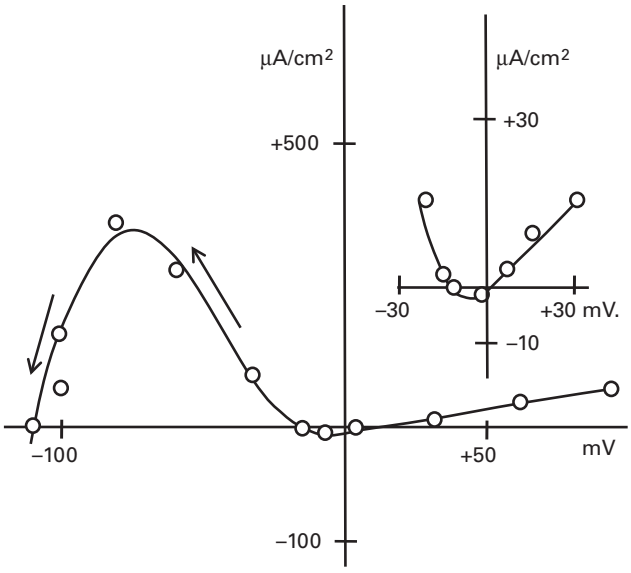


Figure 3.8
Current-voltage (I-V) curve at 0.29 msec after shock is delivered. Note that they flipped their voltage convention. Rest is still at 0 mV. Hyperpolarization is positive, and depolarization is negative. Reproduced from [HHK52] with permission.

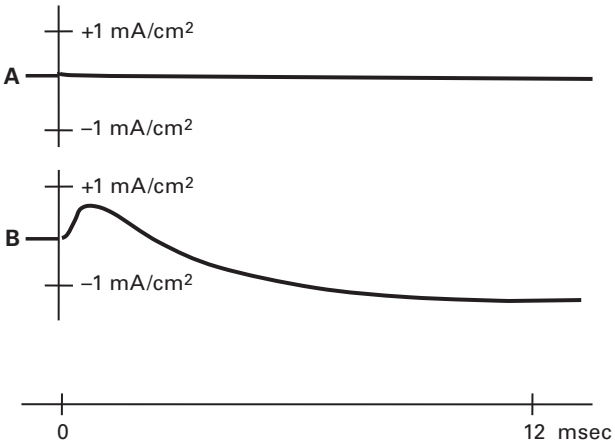


Figure 3.9
Current responses to symmetric positive A and negative B step polarizations. Up in these figures is for inward current, and down from baseline is for outward current. Note the discontinuity at the step onset at 0, which reflects the capacitive current surge not shown on this time frame. Reproduced from [HHK52] with permission.

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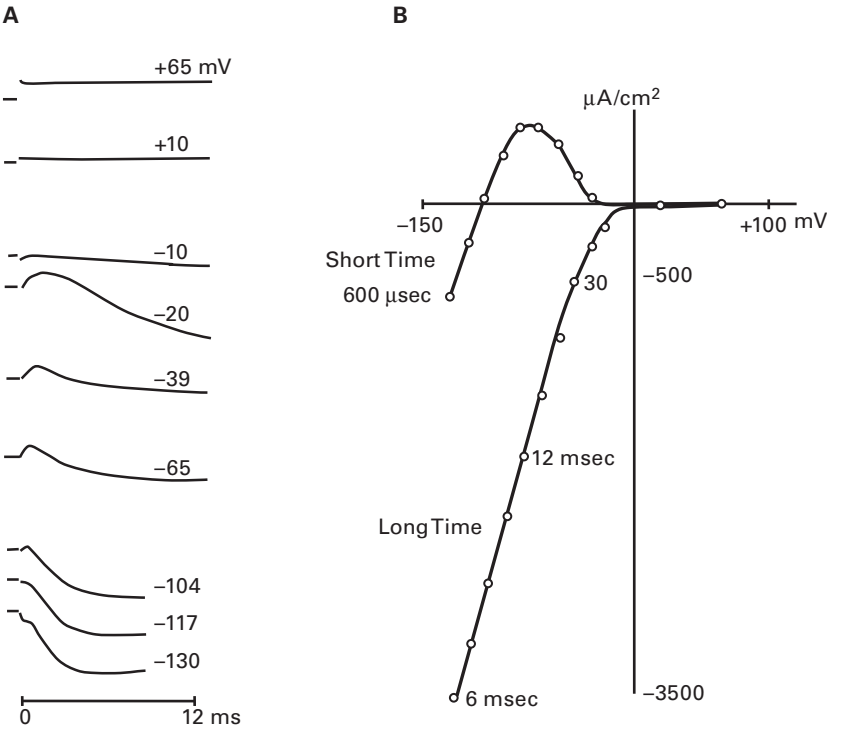


Figure 3.10
A shows currents as a function of time and voltage. B shows current-voltage curves at short versus longer time periods after voltage step. Modified from [HHK52] with permission.

There was a small gap in the records in figure 3.9 at the onset of the change in voltage. If one examines such traces at higher speed, this capacitive current *surge* is seen (figure 3.11). It's important to note that this capacitive current is symmetric, very much unlike the currents in figure 3.9. It is also much faster than the more biologically based mechanisms behind the ionic currents.

3.3 Hodgkin and Huxley

We now move on to the sequence of four dual-authored papers typically referred to as the Hodgkin-Huxley papers.[5]

5. The reader need not be too concerned for the missing colleague Bernard Katz. He would receive the 1970 Nobel Prize for Medicine, seven years after Hodgkin and Huxley.

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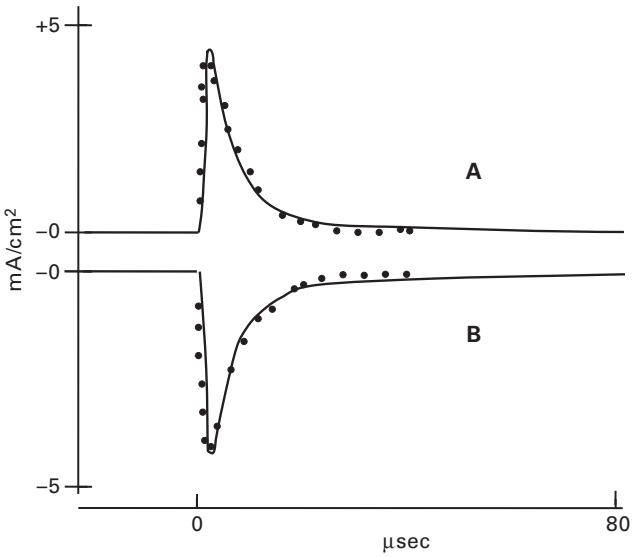


Figure 3.11
Very short time symmetric capacitive currents. Reproduced from [HHK52] with permission.

Hodgkin-Huxley Paper 1

The first task was to separate the ions. The first experiment was to replace, in their artificial seawater, sodium chloride with choline chloride. Choline is a big enough molecule so it's not going to go through the membrane as easily as sodium. Choline rendered squid axons inexcitable, but did not change the resting membrane potential substantially. In choline seawater, the early inward current disappeared (figure 3.12). Replace the sodium and the early inward current reappears. Note the early outward current bump (gray arrow) in figure 3.12. It is due to sodium current flowing from inside to outside as the electrical gradient holding it in is decreased under the depolarizing pulse.

A more complete sequence of hyperpolarizing (positive) and depolarizing (negative) pulses are shown in normal versus choline seawater in figure 3.13. Note that the transient inward sodium current decreases as depolarization becomes more intense. It reverses at about -98 mV, the sodium reversal potential E_{Na} (arrow in figure 3.13). In choline seawater the small outward sodium current is seen at all depolarizations. Replacing sodium restores the original findings.

The origin of the negative resistance (conductance) in figures 3.8 and 3.10 is now apparent. At modest depolarizations, the sodium gradient caused sufficient inward current to overpower the expected outward current from a polarization that decreased the transmembrane potential. The reversal of sodium current at strong depolarizations (figure 3.13) now

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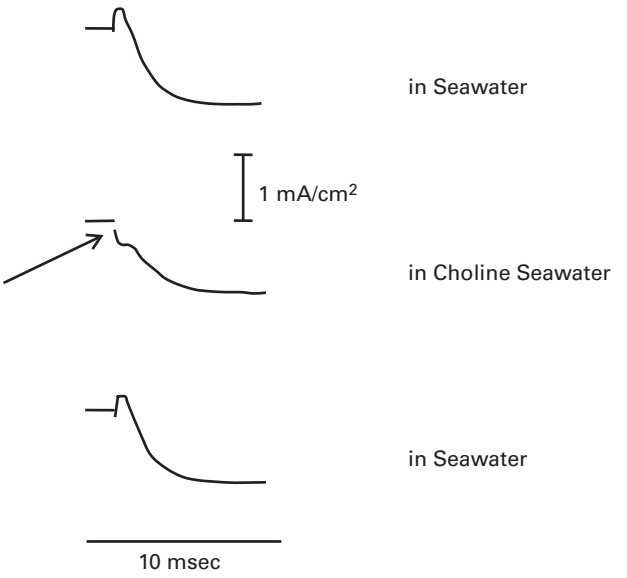


Figure 3.12
Response to 65-mV depolarizations. Modified from [HH52a] with permission.

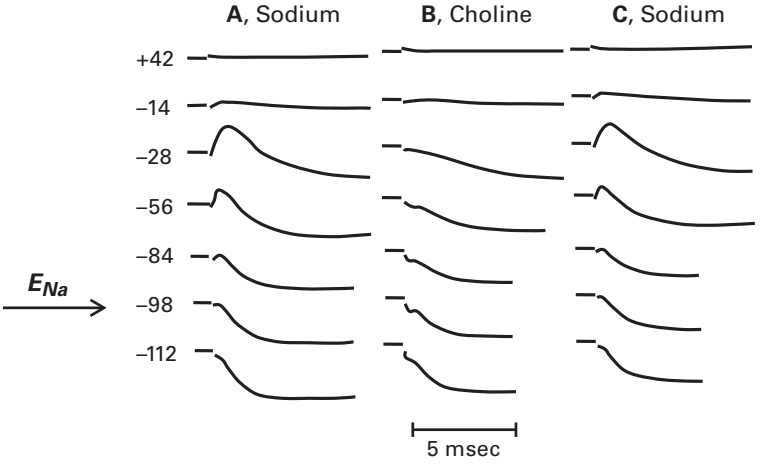


Figure 3.13
Voltage clamp responses in (A) seawater, (B) choline seawater, and (C) seawater. Modified from [HH52a] with permission.

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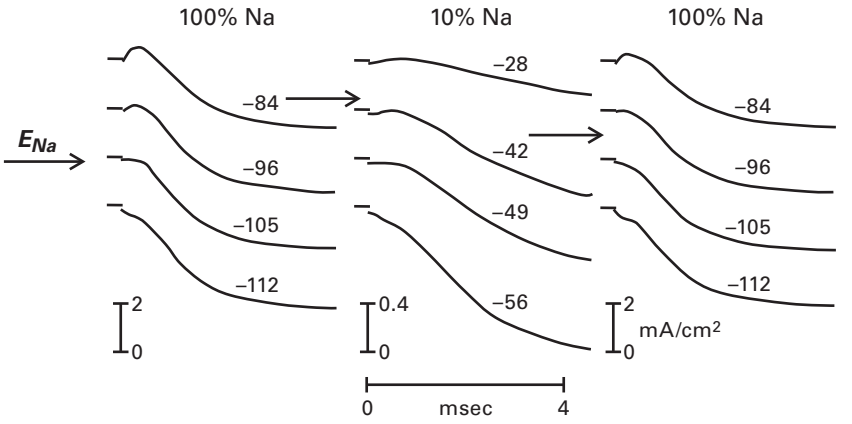


Figure 3.14
Changing external sodium concentration to test whether the Nernst potential accounted for the early outward current's reversal potential. Modified from [HH52a] with permission.

gives you the ability to estimate what the internal sodium concentration is. From the Nernst potential

$$E_{Na} = \frac{RT}{nF} \ln \frac{[Na]_o}{[Na]_i} \approx 26mV \ln \frac{[Na]_o}{[Na]_i} \tag{3.2}$$

where R is the gas constant, n the valence, T temperature in degrees Kelvin, and F the Faraday constant, so that RT/nF is about 26 mV at 25°C. Given the outside sodium concentration $[Na]_o$ of 460 mM in seawater, they estimated the internal $[Na]_i$ to be 60 to 70 mM.[6]

Hodgkin and Huxley did a series of experiments (figure 3.14) to test whether sodium was the early current carrier by varying the sodium concentration, and asking if the change in reversal potential tracked the Nernst potential in equation (3.2). The comparison of theory with experiment was nearly exact. The Nernst potential could be used to calculate at what value to set the voltage clamp in order to reverse the early current.

The early current was therefore a sodium current. It flowed in response to both concentration difference of sodium inside and outside the nerve, as well as the electrical potential gradient. One gradient could overcome the other.

The late current appeared Ohmic (figure 3.10), but it is huge, and much greater (50 to 100 times) than what would be expected from just the voltage change alone. It seemed unaffected by substituting choline for sodium in the external bath. So assume that the total ionic current, $I_i = I_{Na} + I_K$, and in low sodium, the total current, $I_i^{'}$, will be $I_i^{'} = I_{Na}^{'} + I_K^{'}$. If we assume

6. We will explicitly examine mammalian $[Na]_i$ and $[Na]_o$ dynamics in chapter 12.

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that $I_K = I_K^{'}$, then $I_i - I_i^{'} = I_{Na} - I_{Na}^{'}$. It follows that $I_K = I_K^{'} = I_i - I_{Na}^{'}$. Last, assume that I_i and $I_i^{'}$ have the same time course, so that they are related as $k = I_{Na}^{'}/I_{Na}$.

So measure the total current in normal I_i and low sodium $I_i^{'}$ conditions, and one backs out a whole set of relationships to calculate both I_K and I_{Na}

$$
\begin{aligned}
I_i - I_i^{'} &= I_{Na} - I_{Na}^{'} = I_{Na} - I_{Na}\left(\frac{I_{Na}^{'}}{I_{Na}}\right) = I_{Na}(1-k) \\
I_{Na} &= \left(I_i - I_i^{'}\right) \Big/ (1-k) \\
I_{Na}^{'} &= k\left(I_i - I_i^{'}\right) \Big/ (1-k) \\
I_K = I_K^{'} &= I_i - I_{Na} = \left(I_i^{'} - kI_i\right) \Big/ (1-k)
\end{aligned}
\tag{3.3}
$$

Using these relationships, they can now estimate the individual currents as shown in figure 3.15.

Their next assumption was that the driving force for these ion currents was equal to the voltage applied by their voltage clamp, V, minus the Nernst potential for a given ion species, $E_{Na/K}$. $I_{Na/K}$ would then be proportional to the driving force $(V - E_{Na/K})$. If you put the command potential equal to the Nernst potential, no current would flow. The

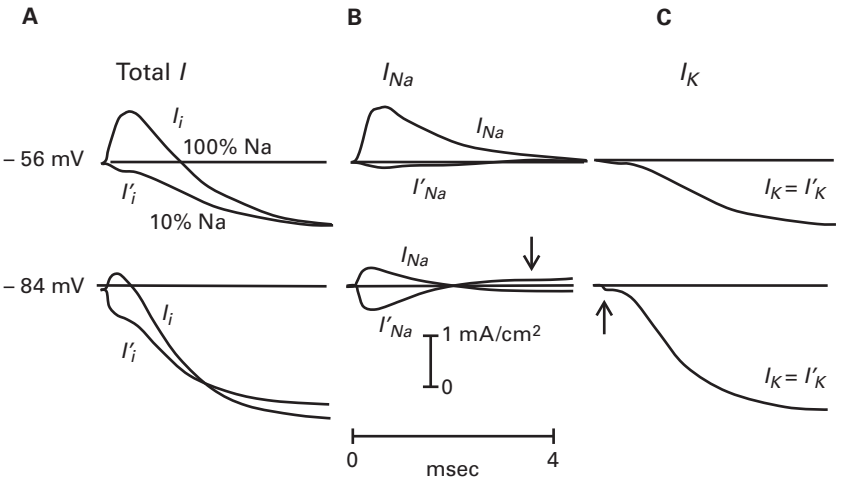


Figure 3.15
Separating currents. Note that near the sodium equilibrium potential in the lower plots, the tails in calculated I_{Na} and $I_{Na}^{'}$ cross (down gray arrow), showing that the assumptions in equations (3.3) were not perfect. Another imperfection is evident in the small offset (up gray arrow) at the beginning of the calculated I_K at -84 mV. This is due to other ions, such as chloride, not included in these assumptions, but which will form the *leak* current to be discussed shortly. Modified from [HH52a] with permission.

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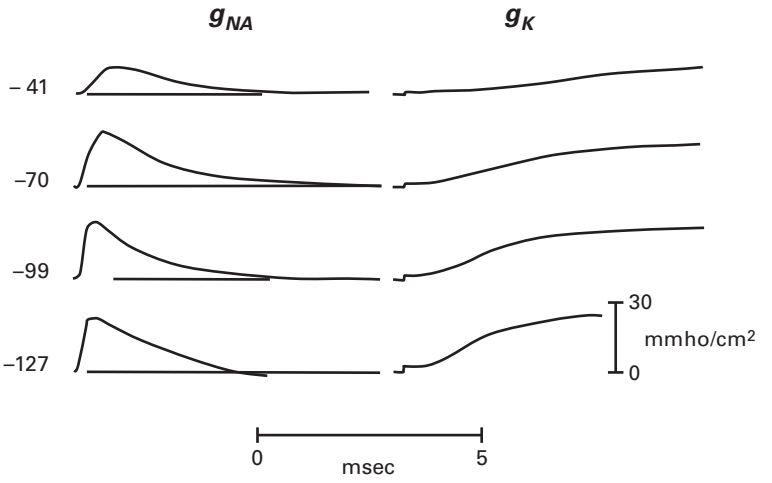


Figure 3.16
Families of conductance curves. Values of E_K are calculated from the subsequent Hodgkin-Huxley paper. Note that mho, Ohm spelled backwards, is an obsolete term for Seimens (1/Ohm). Modified from [HH52a] with permission.

magnitude of the current would then be proportional to the permeability or conductance, g, for a given ion species, or

$$g = I_{Na/K}/(V - E_{Na/K})$$

Calculating families of conductance curves, they obtain figure 3.16. Note that both the maximal conductances and the rate of change in conductance are voltage dependent. These two features are important in what follows.

The increase in conductances from rest were tremendous. Putting together data from the experiments that were sufficient to piece together curves,[7] they obtained the data in figure 3.17. The logarithmic plots demonstrate up to 1,000-fold increases in conductance when depolarizing from rest. These figures were fully consistent with what Cole and Curtis [CC39] found from their experiment in figure 3.1, measuring total conductance increase at the peak of the action potential.

Last, they proposed in this first paper that each ion species crossed the membrane independently of each other. This *independence principle* was a key to working out their model. It will also prove the likely explanation for why, in chapter 12, we will be able to fully reconstruct Hodgkin-Huxley dynamics using a control theoretic approach.

7. There were a whole series of experiments that Hodgkin and Huxley had performed previously. Throughout their four 1952 papers they carefully pick their examples. Many of these axons were also deteriorating, some oscillated, others didn't, but they picked the ones that seemed to give them data that they could use. One gets the impression that if they actually used their data in an unbiased fashion, perhaps with the benefit of a statistician looking over their shoulder reminding them not to throw any of the data away or be biased about what they picked, they would never have actually gotten to the conclusions that they came to.

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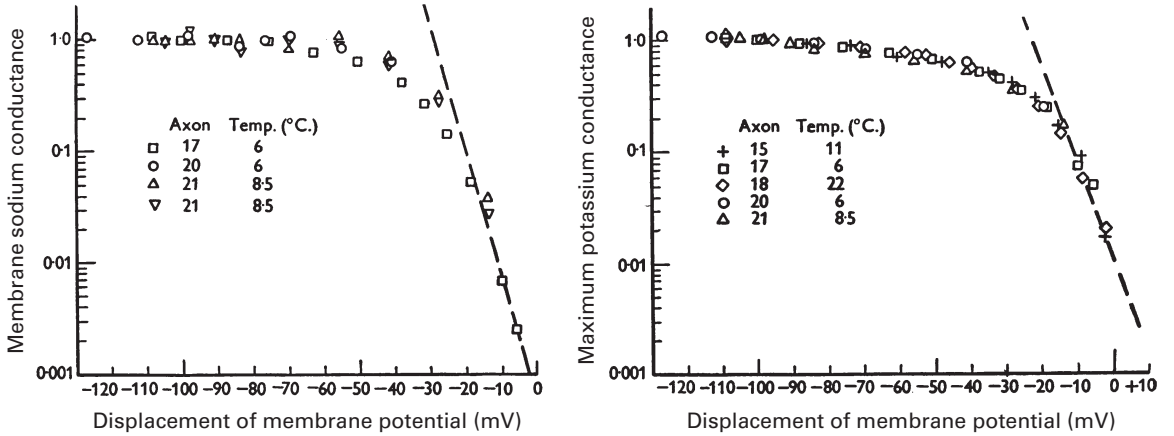


Figure 3.17
Semilogarithmic plots of peak conductances for sodium and potassium from selected axons. Reproduced from [HH52a] with permission.

Hodgkin-Huxley Paper 2

The second concerns general changes in voltage and time. The first experiments used pulses of different time lengths. If they hit the nerve with a short enough pulse—5, 50, maybe 80 μsec—they got nothing. The large inward shift in sodium required that they hold the nerve membrane depolarized for a sufficient length of time for some process to occur. They could then stop it dead in its tracks by cutting the pulse off. The biological mechanism that underlay the inward sodium current was reversible (figure 3.18).

When Hodgkin and Huxley depolarized the membrane to the sodium equilibrium potential, and then repolarized it back to rest, a significant inward current appeared, which exponentially decayed. This current disappeared in choline seawater (figure 3.19).

The discontinuities in current, such as those observed in figures 3.18 and 3.19, disappeared with the calculation of conductance, as shown in figure 3.20. Physically, this makes sense. Change the potential discontinuously and the current changes discontinuously. But conductance is continuous, physically as well as mathematically, and it takes a relatively long time for a biological conductance to change in response to voltage changes. The ratios of the two discontinuous processes tend to cancel each other out in the calculation of conductance. A continuous process would prove essential for them to be able to fit curves in developing their model of these dynamics.

Now depolarize the membrane, and then quickly change to a different voltage. Recall that conductance cannot change instantaneously. So the long time change in current with the first depolarization, shown in the nonlinear curve with the negative resistance region in figure 3.21, is very different from the I-V relationship measured just after the voltage

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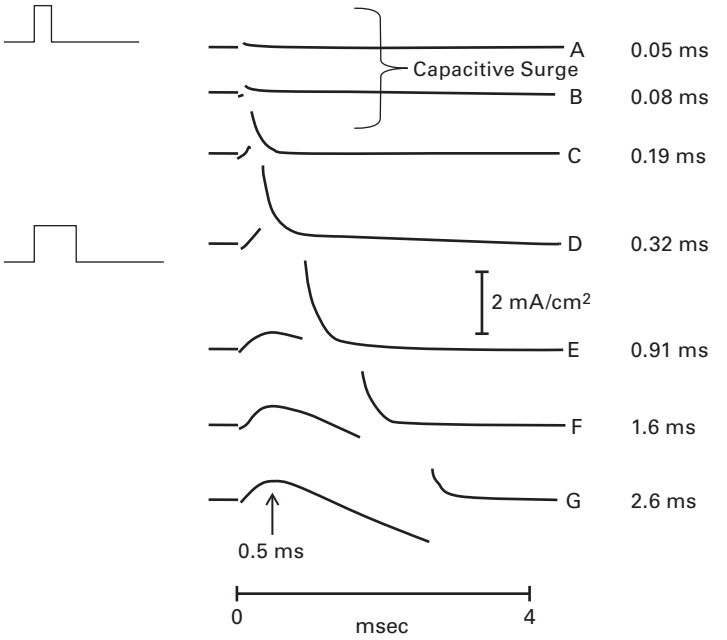


Figure 3.18
Membrane currents with depolarizing pulses of different lengths. Modified from [HH52b] with permission.

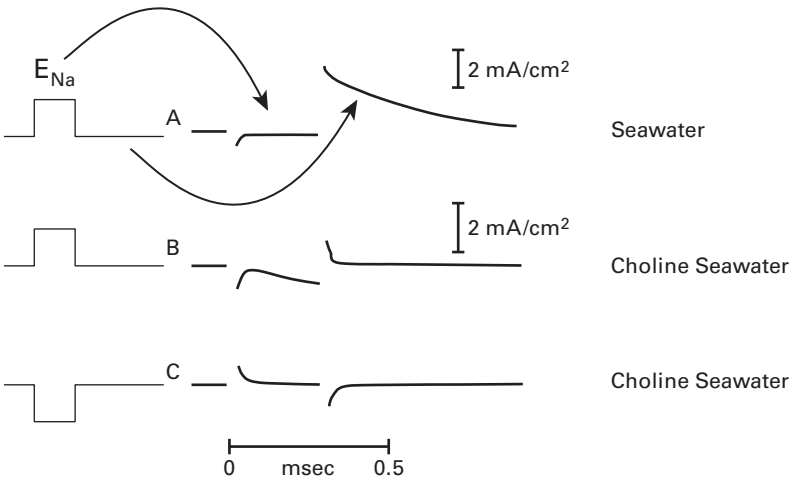


Figure 3.19
Inward sodium current *after* depolarizing pulse. Note the small outward current developing during depolarization in low-sodium external solution. Modified from [HH52b] with permission.

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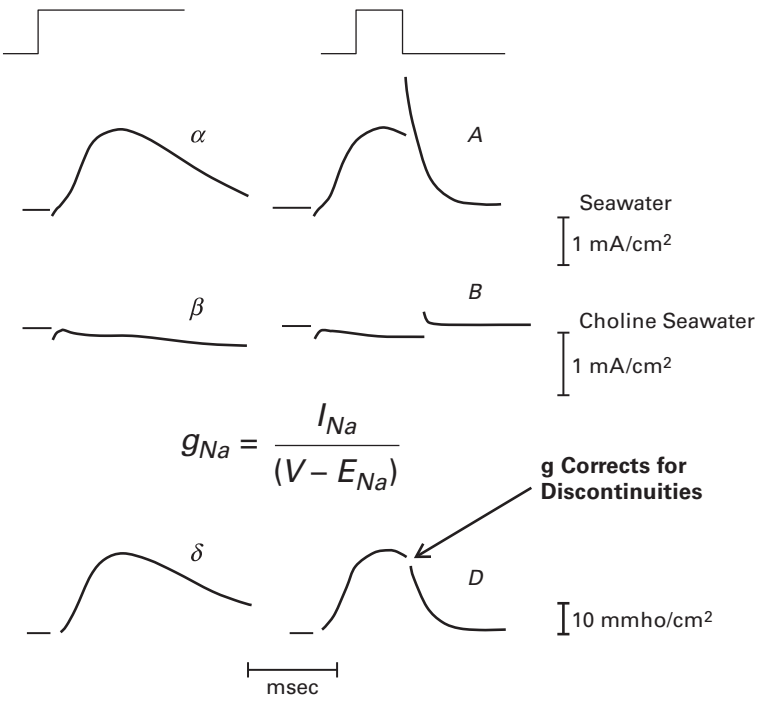


Figure 3.20
Current versus conductance traces for step and square pulse depolarizations. Modified from [HH52b] with permission.

is changed, when a linear relationship is observed. Note that when $V_1 = V_2$, the curves intersect—in this case both tests are the same, a depolarization to V_1.

The next observation is that the rate of decline in g_{Na} is proportional to the value of g_{Na} reached (figure 3.22).

In switching now to potassium currents, Huxley and Hodgkin use choline seawater, so that no sodium contributes to inward current, and after a depolarization, they rapidly repolarize to study the potassium currents.

Just as with sodium currents, they could contrast the nonlinear long-term versus the instantaneous chord conductance measurements for potassium from an instantaneous change in voltage. Since conductance cannot change instantaneously, the instantaneous I-V curve is linear in figure 3.23.

If one plots the turn-on and turn-off of g_K on a linear-linear scale, as in figure 3.24, these conductances are observed to turn on with a sigmoid-shaped function and to decay exponentially. As for g_{Na}, the rate of decay for g_K is proportional to the level of g_K reached (not shown).

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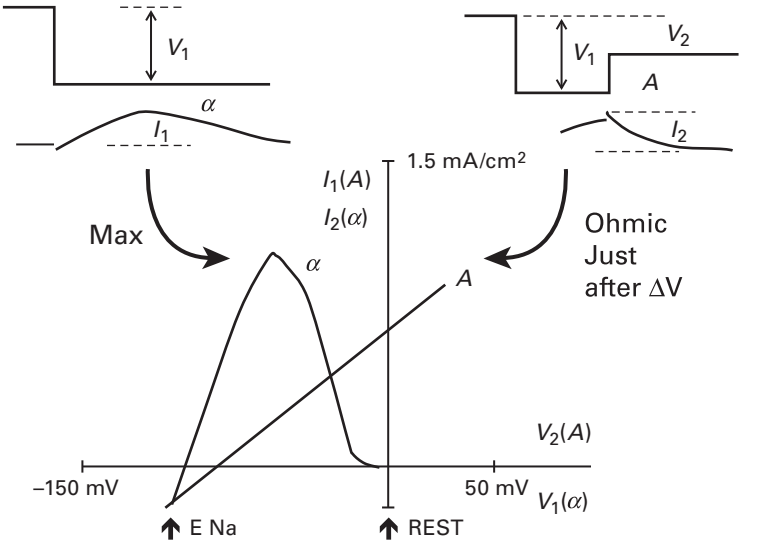


Figure 3.21
Contrast of slow nonlinear response to depolarization and rapid Ohmic response to second voltage change before the conductance can change. Modified from [HH52b] with permission.

Hodgkin-Huxley Paper 3

The third Hodgkin-Huxley paper concerns inactivation. Potassium turns on and stays on; sodium turns on and then off. Some process turns off sodium conductance when the membrane is depolarized. Sorting this out will take some more complex experiments.

Hodgkin and Huxley employed conditioning voltages—small changes in potential followed by large pulses. These small conditioning pulses were not associated with any appreciable inward currents, but they substantially altered the later response of the nerve. An example is shown in figure 3.25. Small conditioning depolarizations will progressively reduce the inward sodium current, but do not affect the subsequent outward potassium current.

A time course of this conditioning effect is evident in figure 3.25. Decreasing the ability of sodium to rush in was termed *inactivation*. It turns out that conditioning with small hyperpolarizing potentials increased sodium current, and this was termed *deinactivation*. By measuring the current produced and varying the conditioning pulse time, one could develop time courses of (de)inactivation and estimate a time constant for this process, as shown in figure 3.26. Deinactivation will be critical to our discussion of model-based control of Parkinson's disease in chapter 10.

If they perform a long enough conditioning pulse, they achieve the *infinite time* inactivation, described through variable h_∞ (or more precisely $1 - h_\infty$). Condition with a long

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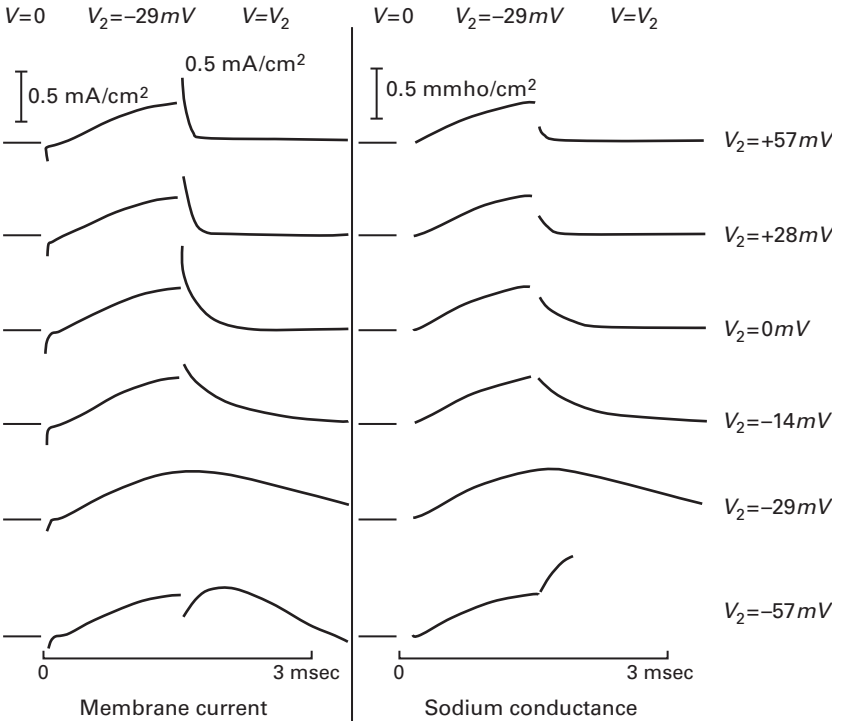


Figure 3.22
The rate of decay of conductance is proportional to the level of conductance reached. Reproduced from [HH52b] with permission.

enough hyperpolarizing pulse, and h_∞ goes to 1. It is as if the membrane is wide open. It fires with maximal sodium current strength. In figure 3.27 h_∞ traces a sigmoid curve. The normal rest potential (unconditioned) sits very near the inflection of the sigmoid (often assumed about $h = 0.6$). One positions systems near such inflection points (whether nerve membrane or transistor) in order to preserve maximum dynamic range and responsiveness of a system.

Last, they will study the recovery of inactivation by using paired pulses as in figure 3.28. If the paired pulses are too close together, there is no response. This property of inactivation is a major contributor to the phenomenon of the *refractory period*. The rate of recovery is very similar to the onset of inactivation—the time constant of these processes is about 12 msec. Both processes are dependent on the membrane potential.

Hodgkin-Huxley Paper 4

At present the thickness and composition of the excitable membrane are unknown. Our experiments are therefore unlikely to give any certain information about the nature of the molecular events underlying

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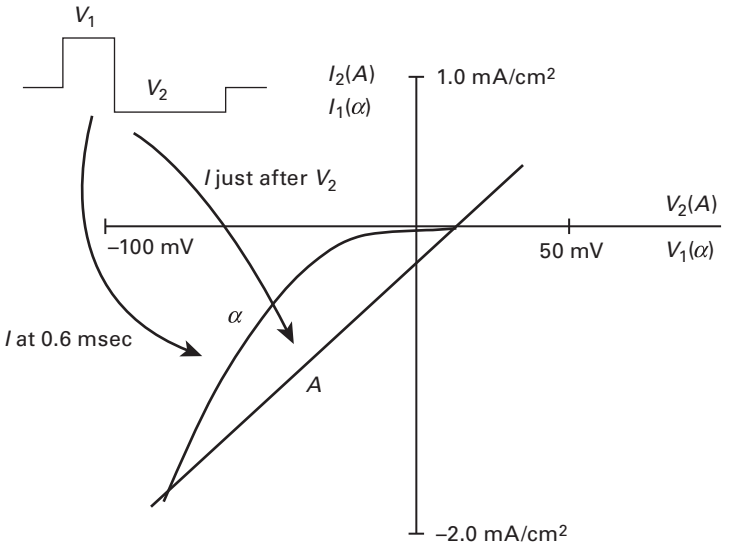


Figure 3.23
Contrast of delayed versus instantaneous I-V curves for potassium. Measured in choline seawater. From [HH52b] with permission.

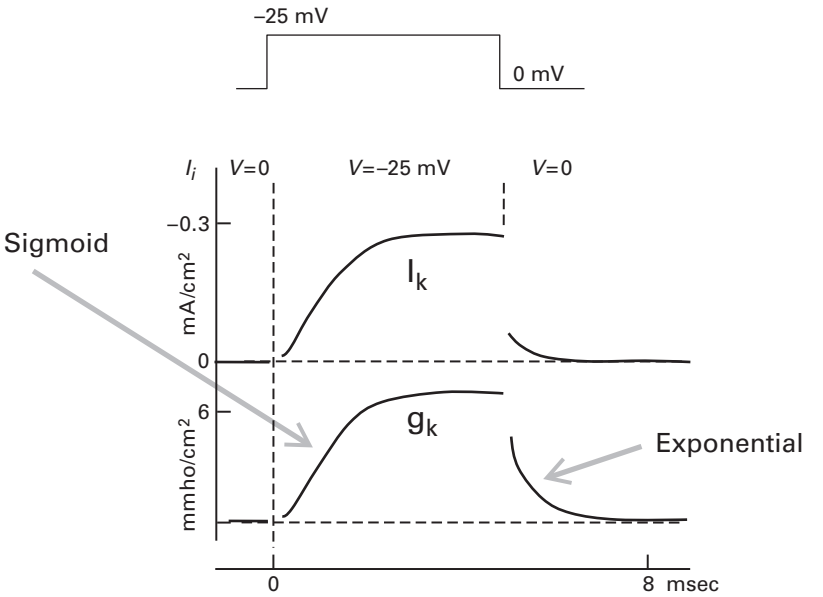


Figure 3.24
Sigmoid turn-on and exponential decay of turn-off of g_K. Modified from [HH52b] with permission.

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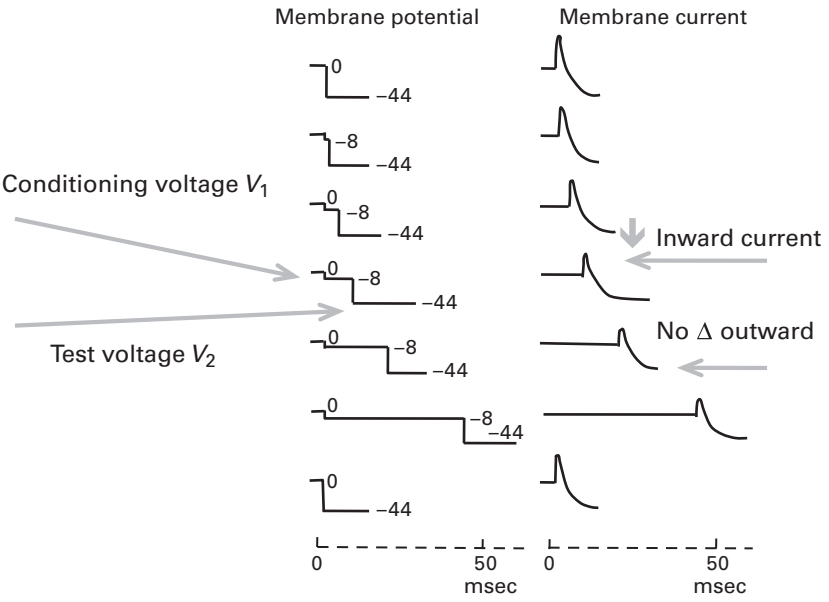


Figure 3.25
Demonstration of conditioning, V_1, and test, V_2, voltages on membrane current. Modified from [HH52c] with permission.

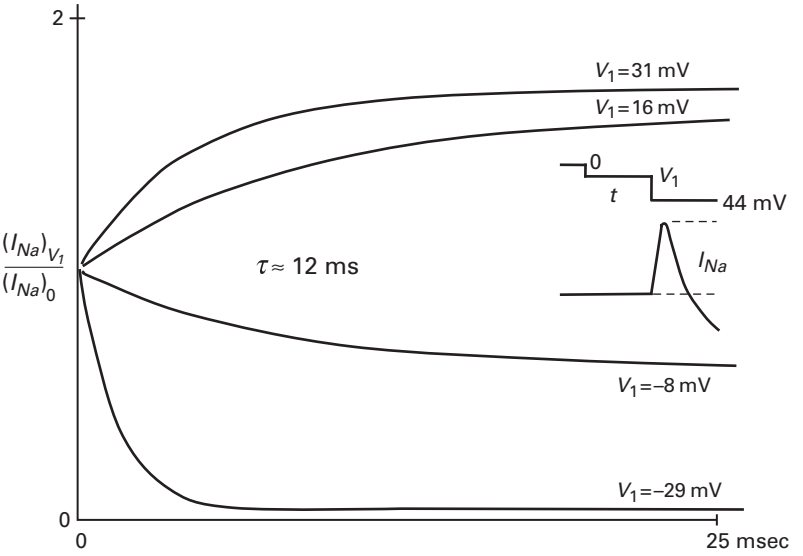


Figure 3.26
Time dependency of inactivation (curves going down) and deinactivation (curves going up). Modified from [HH52c] with permission.

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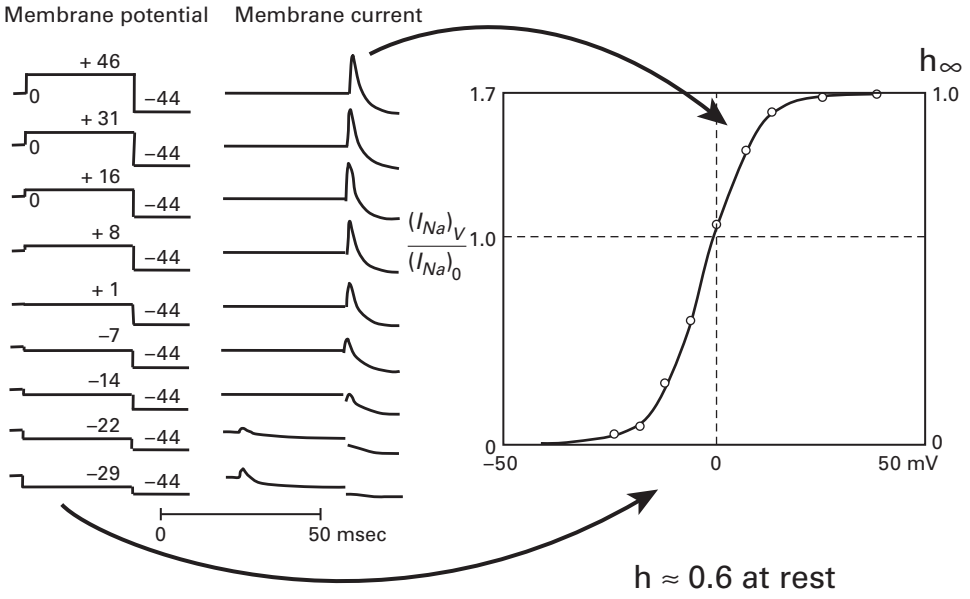


Figure 3.27
Plotting of h_∞. Modified from [HH52c] with permission.

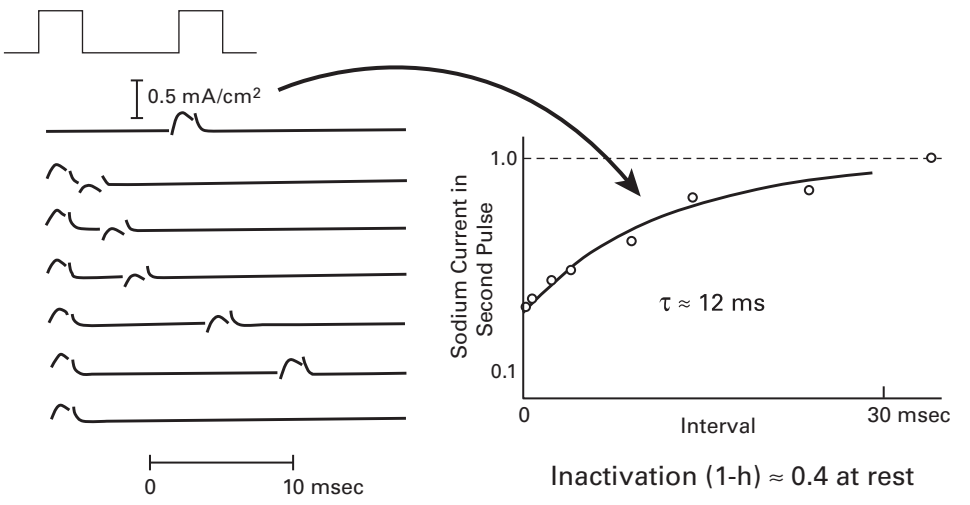


Figure 3.28
Paired pulses to measure recovery from inactivation. Modified from [HH52c] with permission.

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changes in permeability . . . for the sake of illustration we shall try to provide a physical basis for the equations, but must emphasize that the interpretation given is unlikely to provide a correct picture of the membrane. [HH52d]

It is important to understand what these equations represent. Hodgkin and Huxley construct a model that is capable of replicating their data, and will account for a range of phenomena that were known to take place in neurons. What this successful model fitting did not do was explain the underlying molecular events. There are no *channels*, *subunits*, *gates*, or other more modern terminology from molecular biology that influenced their theory. We are about to sketch out a set of deterministic equations that literally come from empirically fitting curves to their data. Therein lies their success and their limitations. Neuronal dynamics are stochastic. Fitting them with deterministic equations can be shown to dramatically fail [MS95]. Taking into account the underlying stochastic dynamics is at times crucial [Fit65] [SFS98], but remains insufficiently appreciated.

To summarize their evidence, Hodgkin and Huxley sketch out the model in circuit formalism that they think their data support (figure 3.29). There are two routes for current to pass—through capacitance or through ionic currents. The conductances for sodium and

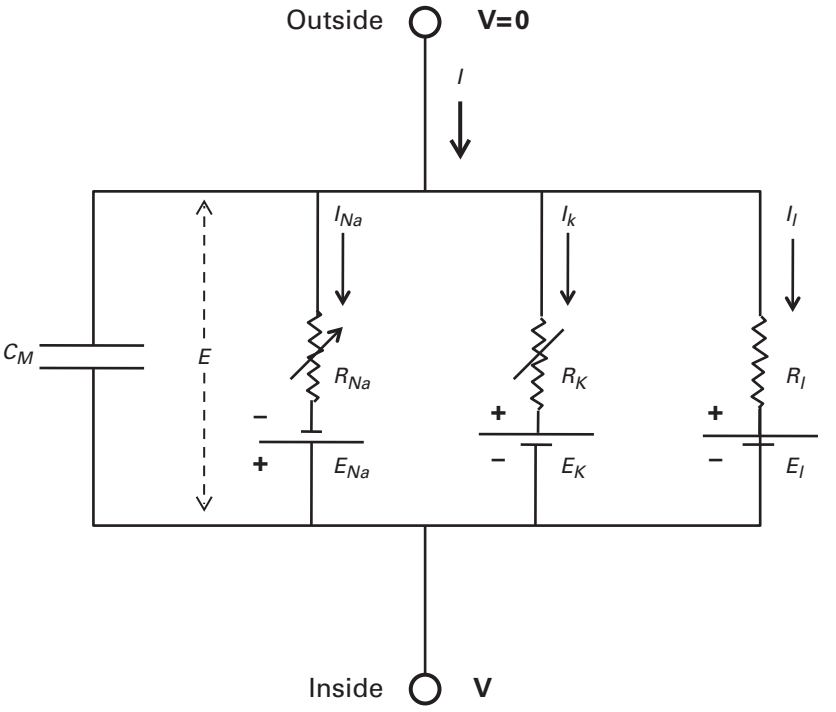


Figure 3.29
Model electrical circuit formalism for axon membrane that experimental data supported. Modified from [HH52d] with permission.

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potassium are variable. The reversal potentials, E_{Na} and E_K, are represented as batteries; they are set by the concentration gradients and calculated through the Nernst potential. Last, there is a *leak* current, where all unaccounted for currents are lumped. Hodgkin and Huxley assume this leak has a conductance that is not voltage dependent, and a constant reversal potential. Later we will discuss *model inadequacy* in some detail (chapter 8). The leak was largely due to chloride, and in practice one could estimate it from the bias current on the voltage clamp at rest. But in a broader sense the leak was also their model inadequacy.

Hodgkin and Huxley assumed from their observations that changes in permeability depended on membrane potential rather than current. They suggested that these permeability changes had something to do with a dipole, a molecule in the membrane that was influenced by the electrical field. They could measure no movement (current) associating with such molecular effects (it would take another twenty years to demonstrate *gating currents* [AB73]). The absence of measurable gating current implied that the sites that controlled the current were sparse in relation to the charge carriers of the ionic current.

Assume, as before, that the total membrane current, I, is the sum of capacitive current, $C_M \frac{dV}{dt}$, and ionic currents, I_i. Furthermore, assume that the total ionic currents are the sum of sodium, I_{Na}, potassium, I_K, and lump the rest into leak current, I_l.

$$I = C_M \frac{dV}{dt} + I_i$$

$$I_i = I_{Na} + I_K + I_l$$

Further assume that the ionic currents are separable, based on the independence principle, into the separate currents

$$I_{Na} = g_{Na}(V - V_{Na})$$

$$I_K = g_K(V - V_K)$$

$$I_l = \bar{g}_l(V - V_l)$$

with variable conductances for sodium and potassium, g_{Na} and g_K, and a fixed conductance for leak, $\bar{g}_l$. The respective equilibrium potentials are written as V_{Na}, V_K, and V_l, where each $V = E - E_r$ is the difference in potential from resting membrane potential.

Recall the sigmoidal rise in g_K shown in figure 3.24, and the simple exponential fall. The simplest chemical system that can produce such dynamics assumes first-order kinetics—that a substance changes only in proportion to its own concentration to another form and back—with rates proportional to α and β[8]

8. Recognize that at the time, there was much prevailing literature that would have suggested a carrier molecule, or some other process interacting with their model of n, in order to transport ions across a membrane. It is one of the remarkable insights of Hodgkin and Huxley to use first-order kinetics as the absolutely simplest model capable of representing their findings. William of Occam would be pleased.

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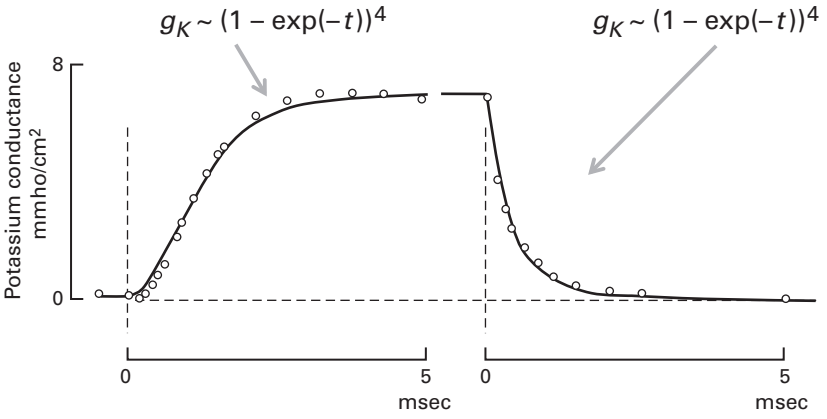


Figure 3.30
Fitting (solid lines) potassium conductance g_K rise and fall to data (circles) from figure 3.24. Modified from [HH52d] with permission.

$$n \underset{\alpha}{\overset{\beta}{\rightleftarrows}} (1-n)$$

Such a process is accounted for by the differential equation

$$\frac{dn}{dt} = \alpha_n(1-n) - \beta_n n \tag{3.4}$$

which literally says that the rate of change in concentration of n in time, dn/dt, is proportional to the *rates* α_n and β_n multiplied by their respective chemical species concentrations n and $1-n$. The sigmoid of the g_K turn-on could be fit by a third or fourth power of such an equation, and Hodgkin and Huxley selected a fourth power fit. Using $g_K \sim (1-\exp(-t))^4$ to fit the increase, and $g_K \sim \exp(-4t)$ for the decay, they achieved the fits shown in figure 3.30 using

$$g_K = \bar{g}_K n^4 \tag{3.5}$$

where $\bar{g}_K$ is the maximum potassium conductance.

At rest, $dn/dt = 0$, so if

$$\frac{dn}{dt} = \alpha_n(1-n) - \beta_n n = 0$$

then

$$\alpha_{n_0}(1-n_0) - \beta_{n_0} n_0 = 0$$

$$n_0 = \frac{\alpha_{n_0}}{\alpha_{n_0} + \beta_{n_0}}$$

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provides a boundary condition. The solution is therefore

$$n = n_\infty - (n_\infty - n_0)\exp(-t/\tau_n) \tag{3.6}$$

where

$$n_\infty = \alpha_n/(\alpha_n + \beta_n)$$

$$\tau_n = 1/(\alpha_n + \beta_n)$$

and $1/(\alpha_n + \beta_n)$ defines the time constant τ_n. In fitting figure 3.30, they used a best τ_n, and realized that fitting the turn-on would have been better with a fifth or sixth power—not practical when computation was laborious with hand calculators. Applying this fit to a family of potassium conductance curves yielded the results in figure 3.31.

By using the value of τ_n that offered the best fits of the data, and estimating the maximal conductance g_K to be about 20% greater than the value at their most depolarized voltage

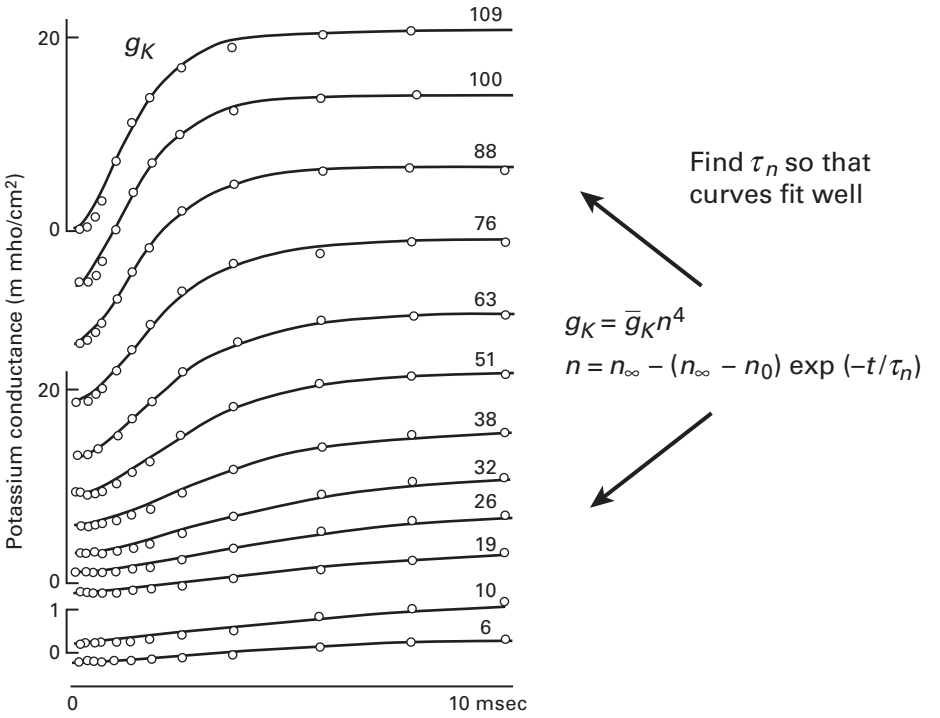


Figure 3.31
Fitting a family of potassium conductance curves with equations (3.5) and (3.6). Modified from [HH52d] with permission.

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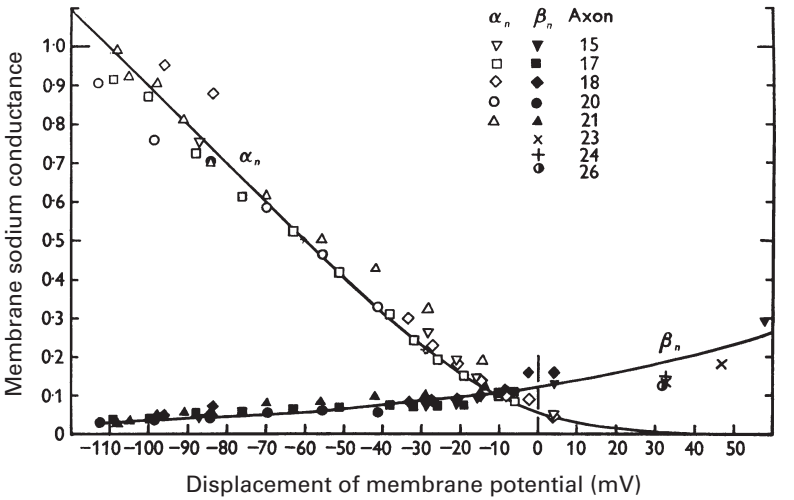


Figure 3.32
The reconstruction of curves for $\alpha_n(V)$ and $\beta_n(V)$. Modified from [HH52d] with permission.

clamp (-110 mV), they could estimate n_∞, and then estimate α_n and β_n from

$$\alpha_n = n_\infty/\tau_n$$

$$\beta_n = (1 - n_\infty)/\tau_n$$

Now all applicable measurement data are collected to piece together functions of $\alpha_n(V)$ and $\beta_n(V)$ as in figure 3.32. The black fit curves in figure 3.32 were empirically fit by

$$\alpha_n = 0.01(V + 10) \Big/ \left[\exp \frac{V + 10}{10} - 1\right]$$

$$\beta_n = 0.125 \exp(V/80)$$

Hodgkin and Huxley struggled with whether to model g_{Na} by a second-order differential equation or assume two variables that obey a first-order equation. They chose the simpler, latter route, assuming that each variable, m and h, obeyed a first-order rate process

$$g_{Na} = m^3 h \bar{g}_{Na}$$

$$\frac{dm}{dt} = \alpha_m(1 - m) - \beta_m m \tag{3.7}$$

$$\frac{dh}{dt} = \alpha_h(1 - h) - \beta_h h$$

As above,

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$$
\begin{aligned}
m &= m_\infty - (m_\infty - m_0)\exp(-t/\tau_m) \\
h &= h_\infty - (h_\infty - h_0)\exp(-t/\tau_h) \\
\tau_m &= 1/(\alpha_m + \beta_m) \\
\tau_h &= 1/(\alpha_h + \beta_h)
\end{aligned}
\tag{3.8}
$$

By experimenting with different ratios of τ_m and τ_h, they came up with the fits in figure 3.33. Using fit values of τ_m and τ_h, they could back out values of α_m, β_m, α_h, and β_h, as shown in figure 3.34. and again from curve fitting, they found

$$\alpha_m = 0.1(V+25) \Big/ \left[\exp\frac{V+25}{10} - 1\right]$$

$$\beta_m = 4\exp(V/18)$$

and

$$\alpha_h = 0.07\exp(V/20)$$

$$\beta_h = 1 \Big/ \left(\exp\frac{V+30}{10} + 1\right)$$

The preceding is the process by which the Hodgkin-Huxley equations were generated. We summarize them here as follows:

$$I = C_M\frac{dV}{dt} + \bar{g}_K n^4(V - V_K) + \bar{g}_{Na} m^3 h(V - V_{Na}) + \bar{g}_l(V - V_l)$$

$$
\begin{aligned}
dn/dt &= \alpha_n(1-n) - \beta_n n \\
dm/dt &= \alpha_m(1-m) - \beta_m m \\
dh/dt &= \alpha_h(1-h) - \beta_h h \\
\alpha_n &= 0.01(V+10) \Big/ \left[\exp\frac{V+10}{10} - 1\right] \\
\beta_n &= 0.125\exp(V/80) \\
\alpha_m &= 0.1(V+25) \Big/ \left[\exp\frac{V+25}{10} - 1\right] \\
\beta_m &= 4\exp(V/18) \\
\alpha_h &= 0.07\exp(V/20) \\
\beta_h &= 1 \Big/ \left(\exp\frac{V+30}{10} + 1\right)
\end{aligned}
\tag{3.9}
$$

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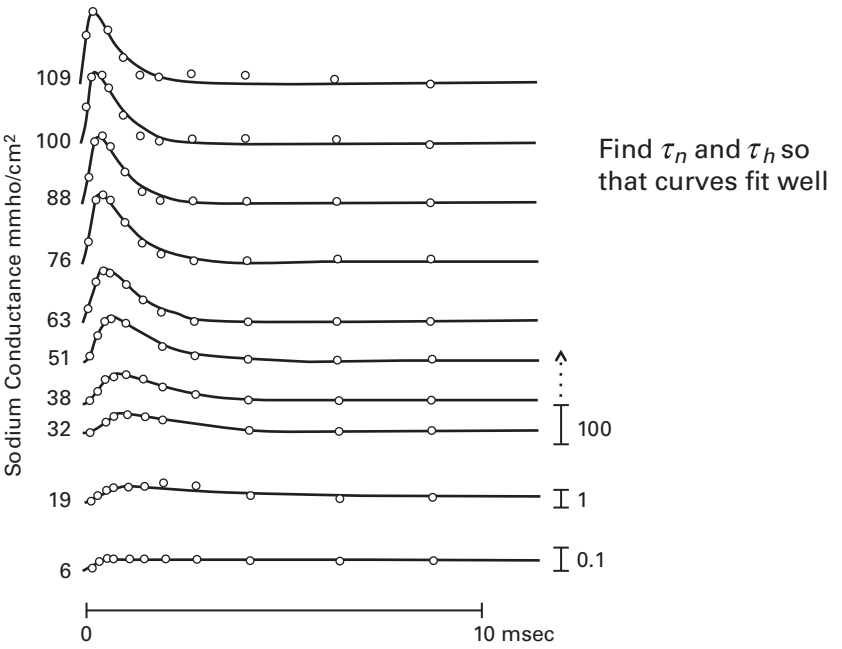


Figure 3.33
Fitting a family of sodium conductance curves with equations (3.7) and (3.8). Modified from [HH52d] with permission.

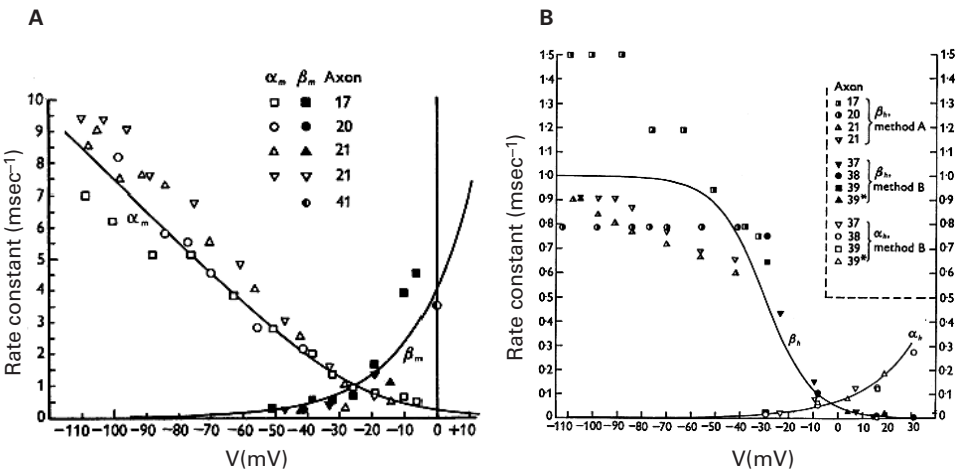


Figure 3.34
The reconstruction of curves (A) for $\alpha_m(V)$ and $\beta_m(V)$, and (B) for $\alpha_h(V)$ and $\beta_h(V)$. Modified from [HH52d] with permission.

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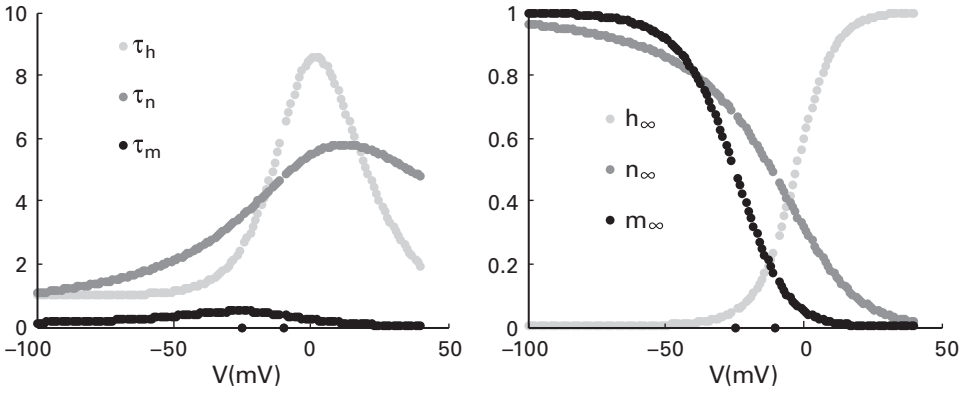


Figure 3.35
Relationship of the time constants and infinite time gate variables.

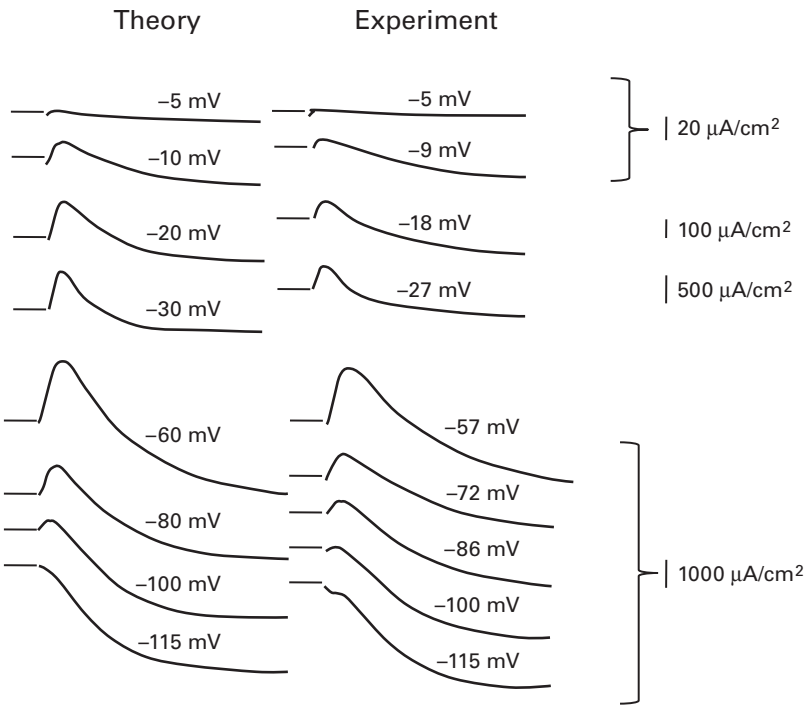


Figure 3.36
Predicting voltage clamp experiments from equations derived from voltage clamp data. Modified from [HH52d] with permission.

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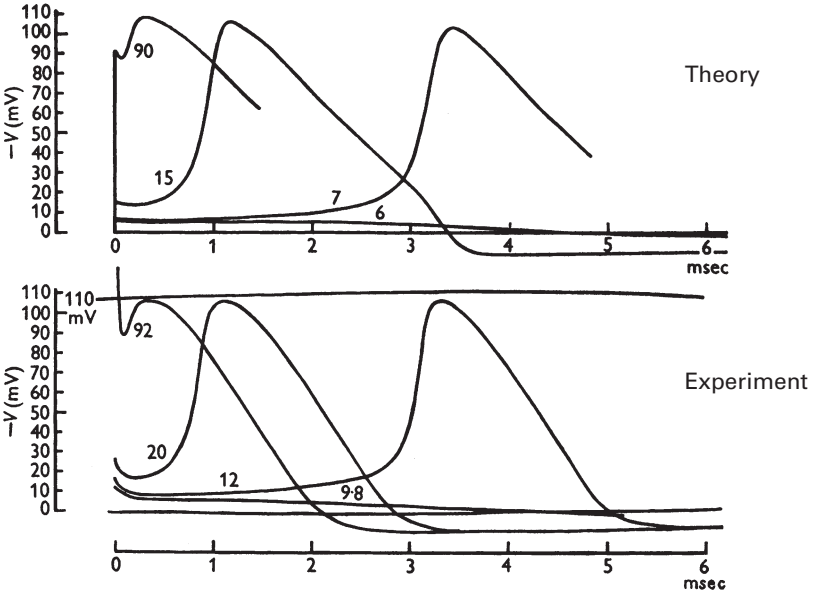


Figure 3.37
Predicting membrane action potential experiments from equations derived from voltage clamp data. Modified from [HHK52] with permission.

Let's start by gathering some intuition regarding the time constants and the long-time behavior of the rate constants.

It is useful to understand that the time constants and infinite time rate constants are functions of voltage, and the relative magnitudes of these time constants and infinite time gate variables are shown in figure 3.35. A small magnitude of τ_i indicates that it is fast. τ_m is fastest. You knew that. Note that τ_h is slowest near rest.

Now let's see if the equations will predict voltage clamp experiments that they were based on. Figure 3.36 shows that they do.

Next go back to the experiments of Hodgkin, Huxley, and Katz (figure 3.6), where they examined membrane action potentials that were *not* under voltage clamp. The replication in figure 3.37 is famous for having only one complete action potential.[9]

Calculating a propagating action potential was much more difficult than calculating the membrane action potential. In essence, one needs to solve a wave equation, by shrinking the membrane action potential (ordinary differential equation) to infinitesimally small elements, and linking these elements together to obtain the variations in space and time (partial

9. Numerical integration using a hand adding machine is laborious. Once they had one complete action potential, and the beginnings of the others, it was sufficient proof of replication.

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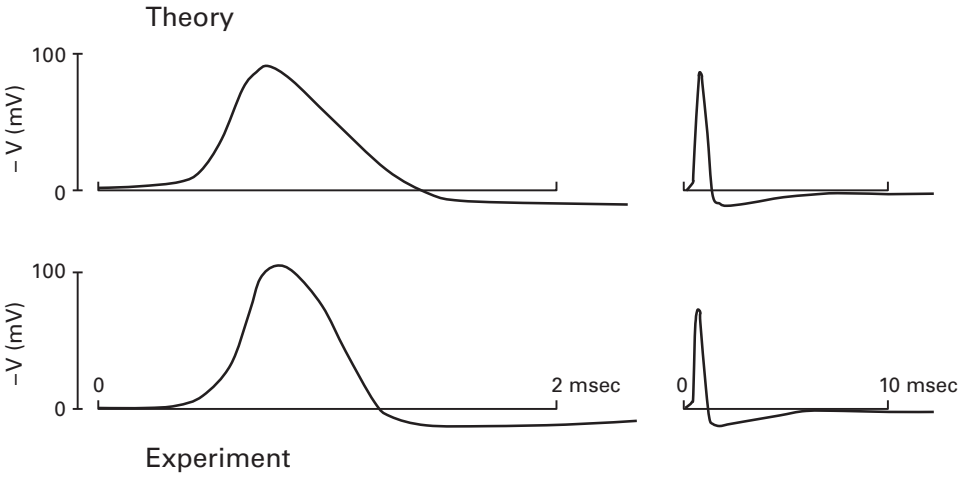


Figure 3.38
Propagated action potential. Modified from [HH52d] with permission.

differential equation). Their numerical integration of a propagated action potential is shown in figure 3.38.

Hodgkin and Huxley summarized their conductances versus voltage for their propagating action potential in figure 3.39, and compared this with Cole and Curtis's plot of the conductance change relative to the propagating action potential shown previously in figure 3.1. The qualitative relationship of total conductance, the sum of g_{Na} and g_K, were in excellent agreement with these previous experiments.

They explained the refractory period following a spike as a combination of inactivation and increased potassium conductance, as shown in figure 3.40. Comparing theory and experiment, the refractory period (relative and absolute) are both replicated in figure 3.41.

In examining the threshold region, note that the subthreshold regenerative response in the experimental traces at 11.8 to 12.0 mA are replicated in the theoretical curve 6 in figure 3.42. We will discuss this threshold behavior in more detail shortly.

One of the strange phenomena of neurons is that if you hold them at hyperpolarized potentials, and then release them quickly back to rest potential, they can exhibit a spike known as *anode break excitation*, which, indeed, the Hodgkin-Huxley model replicates in figure 3.43. They could now explain this phenomenon. Hyperpolarization decreases g_K and inactivation; when released, because the time constants shown in figure 3.35 show that neither n nor h can move quickly, g_{Na} is dominant, permitting an action potential to form. The essence of threshold is the region where inward current (sodium) is greater than outward (potassium), and the rate of change of g_{Na} increasing is greater than that of g_K increasing. This is what happens at the onset of anode break excitation. We will

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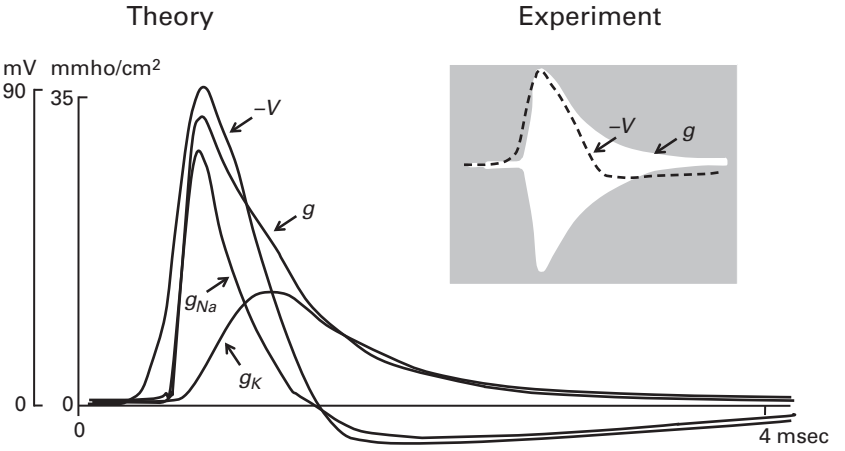


Figure 3.39
Conductance versus voltage for propagated action potential. Modified from [HH52d] with permission.

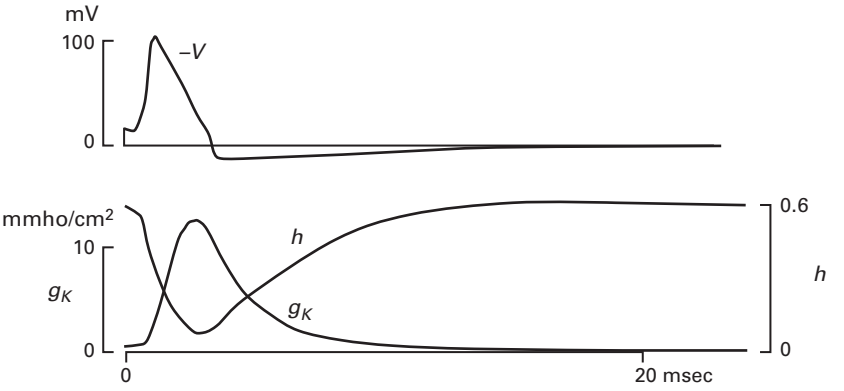


Figure 3.40
Origin of the refractory period is a combination of g_K and h. Modified from [HH52d] with permission.

return to this phenomenon of rebound activation of neurons in chapter 10 in our discussion of Parkinson's disease modeling and control.

Neurons *accommodate*; if you depolarize them too slowly, they will not fire. This is now explained. If the current depolarizes too slowly, then inactivation has a chance to override the increase in m, as does the increase in n.

Last, neural membranes oscillate. Indeed, Hodgkin-Huxley demonstrated that their equations could produce damped oscillations in response to a small short shock with high fidelity to the actual neuronal membranes.

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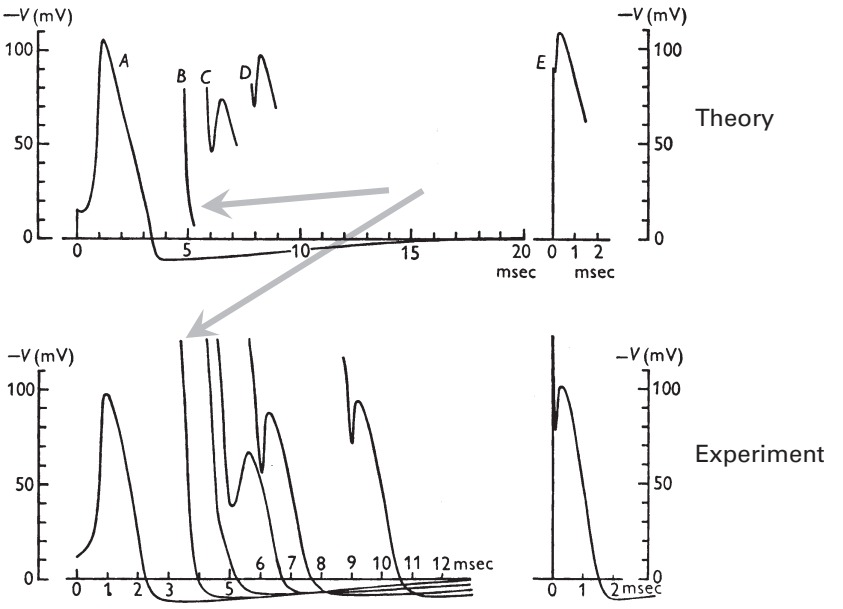


Figure 3.41
Absolute (arrows) and relative refractory period in theory and experiment. Modified from [HH52d] with permission.

The agreement must not be taken as evidence that our equations are anything more than an empirical description of the time-course of the changes in permeability to sodium and potassium ... certain features of our equations were capable of a physical interpretation, but the success of the equations is no evidence in favour of the mechanism of permeability change that we tentatively had in mind when formulating them. [HH52d]

With the perspective of hindsight, their caution was prophetic. Many things were missing from the Hodgkin-Huxley equations. The phenomena of membrane ion *pumps* were not considered. Indeed, we will later discuss the dynamical implications of ion pumps for tracking and controlling phenomena such as epileptic seizures in considerable detail (chapter 12). The *zoo* of ion channels that rain considerable complexity on mammalian neurons was absent [GK02]. Also missing was the extraordinary geometrical complexity of neurons beyond the giant squid axon [Asc02]. And the physics of macromolecular mechanisms [MWC65] that lead to an understanding of stochastic excitability effects on the neuronal membrane had yet to be considered [Fit65] [SFS98].[10]

10. A very nice alternative perspective on this history, with a more mathematical point of view, is John Rinzel's [Rin90]

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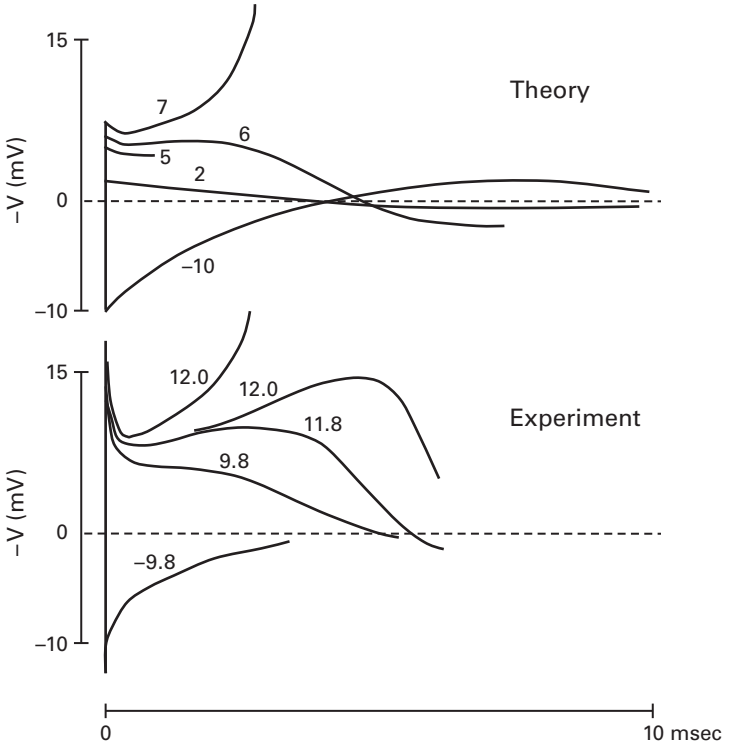


Figure 3.42
Subthreshold regenerative responses in theory and experiment. Modified from [HH52d] with permission.

All of this exploration of the nature of the excitable neuronal membrane is a prelude to reducing the complexity of the Hodgkin-Huxley equations (3.9), in the next chapter, to something more manageable: the Fitzhugh-Nagumo equations. The Fitzhugh-Nagumo equations provide the link to nonlinear control theory in chapter 5 [VTK04]. And all of this will lead us full circle, back to the Hodgkin-Huxley equations and their modifications in chapters 10 and 12.

Exercises

3.1. Prove the solution in equation (3.6) given (3.4).

3.2. These exercises will now guide you through writing your own integrator for the Hodgkin-Huxley equations.

We begin with their original formulation. Use the following:

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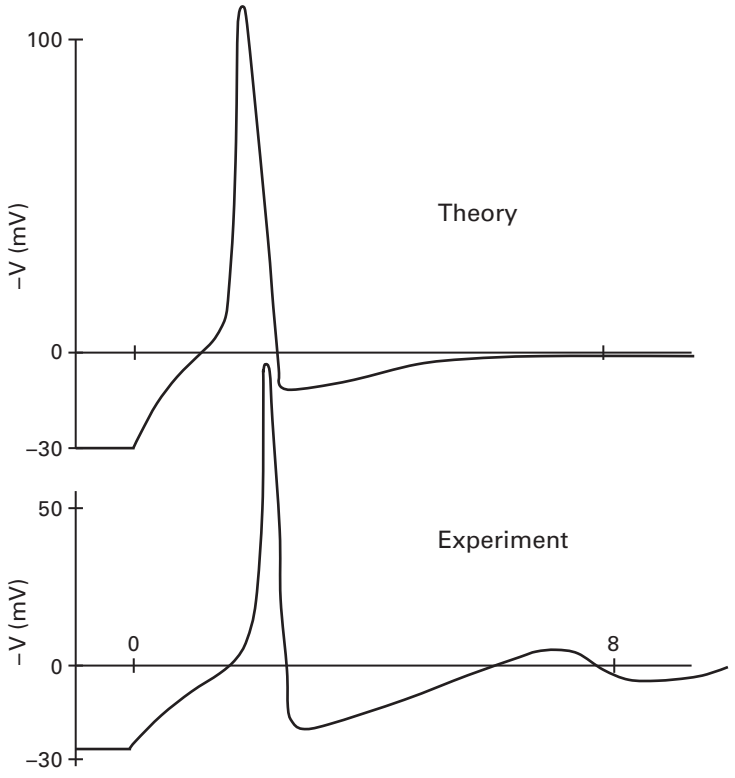


Figure 3.43
Anode break excitation in theory and experiment. Modified from [HH52d] with permission.

- Integration step time of $dT = 0.01$
- Initialize with t = 1, I(t) = 0, and V(t) = 0
- Pulse width of 15 out of 100 integration steps, starting at t = 2

$V_{Na} = 115$

$V_K = -12$

$V_L = 10.6$

$g_{Na} = 120$

$g_K = 36$

$g_L = 0.3$

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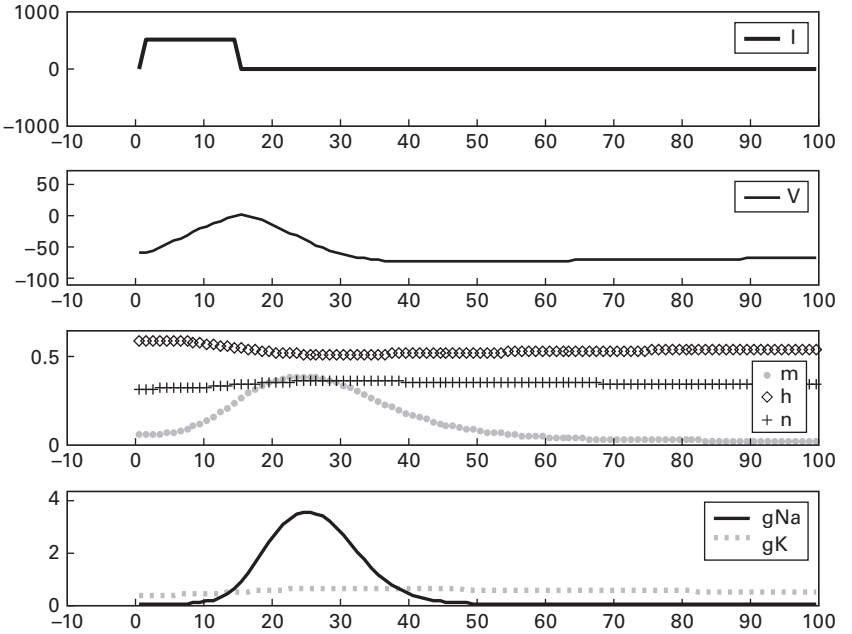


Figure 3.44
Sample output from integrating the Hodgkin-Huxley equations using the preceding suggestions. Pulse amplitude 500 for fifteen time steps.

• Initialize the rate constants, using the formulas from (3.9),[11] and then initialize the gating variables using three formulas as

$$q(1) = a_q(1)/(a_q(1) + B_q(1))$$

where

$$q = m, n, \text{ or } h.$$

• Initialize the conductances

$$g_{Na}(t) = g_{Na} m(t)^3 h(t)$$

and

$$g_K(t) = g_K n(t)^4$$

11. The reader will question what might happen if the voltage approached the singularities of these rate constants, causing them to diverge to infinity. In practice, this rarely happens, and one can go most of a career ignoring this possibility. Perhaps a more cautious approach is to Taylor expand the rate constants near these indeterminate points [MSS82].

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• You will need to make a loop that uses an integrator. Let's start with Euler integration.[12] We wish to evaluate $dy(t)/dt = f(t, y(t))$, using the first two terms of the Taylor series expansion of the function about a point t:

$$y(t+1) = y(t) + \frac{dy(t)}{dt}((t+1) - (t))$$

• We are not under voltage clamp, so the total membrane current is

$$I_m = C_m \frac{dV}{dt} + I_i$$

and if $I_m = 0$, then $C_m \frac{\partial V}{\partial t} = -\sum i_i$. The $\sum i_i$ includes your ionic currents, but also the added ionic current you impose from your current pulse. Write

$$dV = dT * (1/C_M) * \left(-I_{pulse} - \sum I_i\right)$$

and

$$V(t) = V(t-1) + dV$$

• Now update the rate constants and integrate the gating variables as

$$dq = dT(\alpha_q(1 - q(t-1)) - \beta_q q(t-1))$$

$$q(t) = q(t-1) + dq$$

where $q = m$, n, or h.

• If you want to plot with the modern convention, use $V = -(V + 60)$.

• Plot your results for current pulses of from -500 to $+500$. Where is the threshold?[13]

• If you have done this correctly, your results should look similar to figure 3.44:

3.3. Now that you have written an integrator, plot the relationships between the time constants and infinite time rate constants as in figure 3.35.

12. Using Euler integration like this is numerically abysmal [Pre07], but it is the place to start. There is actually a literature on the integration of the Hodgkin-Huxley equations, and the interested reader may wish to review this [MR74].

13. The threshold in the Hodgkin-Huxley equations is considered a *soft* one. We will discuss this in greater detail in the following chapter.

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